

LOAD BALANCING

- ① m servers
- ② n jobs ($0..n-1$), $n \gg m$

FAIR LOAD: $\frac{n}{m}$ jobs per server

GOAL

- Transparency (open source)
- Persistence (job i always go to a small group of servers)
- fair load $\frac{n}{m}$

(round robin creates DoS)

Randomly choose $h \in \mathcal{H}$ (universal), so that
job i has server $h(i)$

$X_j = [\text{Load for machine } j] = [\# \text{ jobs assigned to } j]$

$$X_{ji} = \begin{cases} 1 & \text{if job } i \text{ is assigned to server } j = h(i) \\ 0 & \text{o.w.} \end{cases} \quad \text{pr} = \frac{1}{m}$$

$$X_j = \sum_{i=0}^{n-1} X_{ji} \Rightarrow E[X_j] = \sum_{i=0}^{n-1} E[X_{ji}] = \sum_{i=0}^{n-1} \underbrace{\text{Pr}[X_{ji}=1]}_{\frac{1}{m}} = \frac{n}{m} \quad \text{FAIR LOAD}$$

What about the max load?

$$X = \max_j X_j$$

CONCENTRATION BOUNDS : how close to expectation

- Markov's Inequality (MI)

$$\Pr[X \geq a] \leq \frac{E[X]}{a} \quad (\text{here, we need } a = E[X])$$

- Chebyshev's inequality (CI)

$$\sigma^2 = E[X^2] - E[X]^2 \quad \text{variance}$$

$$\Pr[|X - E[X]| \geq k\sigma] \leq \frac{1}{k^2}$$

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$$X_j = \sum_{i=0}^{n-1} X_{ji}, \quad X_{ji}^2 = X_{ji} \in \{0, 1\}$$

$$\sigma^2 = E[X_j^2] - E[X_j]^2$$

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$$\hookrightarrow \sum_{i=0}^{n-1} \left(\underbrace{E[X_{ji}^2]}_{E[X_{ji}]} - \underbrace{E[X_{ji}]^2}_{\left(\frac{1}{m}\right)^2} \right) = n \left(\frac{1}{m} - \frac{1}{m^2} \right) \quad \sigma = \sqrt{n \left(\frac{1}{m} - \frac{1}{m^2} \right)}$$

$$k = \sqrt{2m}$$

$$\Pr\left[\left|X_j - \frac{n}{m}\right| \geq \sqrt{2m} \sigma\right] \leq \frac{1}{2m}$$

$$\sqrt{2m} \cdot \sigma = \sqrt{2m \cdot n \left(\frac{1}{m} - \frac{1}{m^2} \right)} \sim \sqrt{2n}$$

$$\left. \begin{array}{l} \Pr\left[\left|X_j - \frac{n}{m}\right| \geq \sqrt{2n}\right] \leq \frac{1}{2m} \quad (*) \\ \text{server } j \end{array} \right\}$$

NOTE

$$1) \Pr[X_{ji}=1 \wedge X_{ji'}=1] = \Pr[X_{ji}=1] \cdot \Pr[X_{ji'}=1]$$

$$= \Pr[h(i)=j] \cdot \Pr[h(i')=j]$$

as $h \in \mathcal{H}$ is 2-way independent

2) $X+Y+Z$ pairwise independent $\Rightarrow$

$$\sigma^2(X+Y+Z) = \sigma^2(X) + \sigma^2(Y) + \sigma^2(Z)$$

$$D+2) \Rightarrow \sigma^2(X_J) = \sum_{i=0}^{n-1} \sigma^2(X_{ji})$$

see
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⑥

UNION BOUND

$$\begin{aligned} P_v \left[\max_j \left| X_j - \frac{n}{m} \right| < \sqrt{2n} \right] &= 1 - P_v \left[\exists j : \left| X_j - \frac{n}{m} \right| \geq \sqrt{2n} \right] \\ &= 1 - \sum_{j=0}^{m-1} \underbrace{CI}_{P_v \leq \frac{1}{2m}} \geq \frac{1}{2} \end{aligned}$$



$$\leq \frac{1}{2}$$

In words, the max load is at most $\frac{n}{m} + \sqrt{2n}$ with $p_v \geq \frac{1}{2}$

- Chernoff's bounds (CB)

set of i.i.d variables $y_i \in [0, 1]$, $Y = \sum_{i=1}^n y_i$, $\mu = E[Y]$
*independent
identically
distributed*

► $\Pr[Y > \mu + \lambda] \leq e^{-\frac{\lambda^2}{2\mu + \lambda}}$
*our slack
parameter*

► $\Pr[Y < \mu - \lambda] \leq e^{-\frac{\lambda^2}{3\mu}}$

$$\Pr[Y > \mu + \lambda] \leq e^{-\frac{\lambda^2}{2\mu + \lambda}}$$

hp. $n = m$: $\mu = E[X_j] = \frac{n}{m} = 1$

$$Y = X_j$$

$$\lambda = 6 \ln n$$

$$(*) \Pr[X_j > \frac{n}{m} + 6 \ln n] \leq e^{-\frac{(6 \ln n)^2}{2 \cdot \frac{n}{m} + 6 \ln n}} \leq e^{-3 \ln n} = \frac{1}{n^3} \text{ w.l.p.}$$

$$\frac{-(6 \ln n)^2}{2 \cdot 1 + 6 \ln n} \leq \frac{-(6 \ln n)^2}{2 \cdot 6 \ln n} = -3 \ln n$$

$$\Pr\left[\max_j X_j \leq \frac{n}{m} + 6 \ln n\right] = 1 - \sum_{j=1}^m (*) \geq 1 - n \cdot \frac{1}{n^3} = 1 - \frac{1}{n^2}$$